

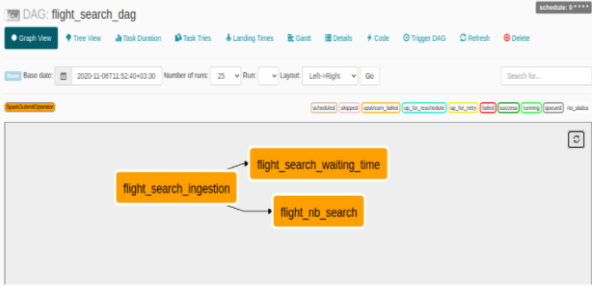
CENTER FOR ADVANCED COMPUTING
PGDBDA Sample Paper-1 (Big Data Technologies)

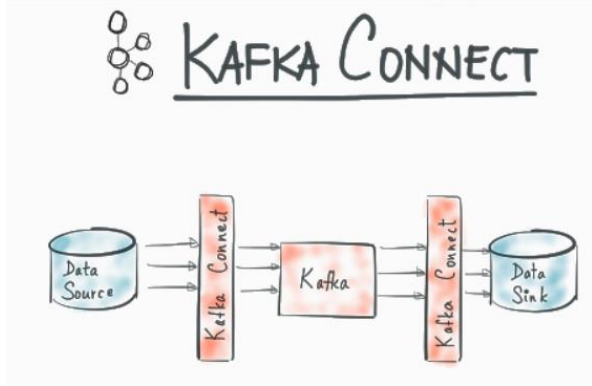
Q.No.	ANSWER ALL THE MCQ QUESTIONS	Marks
1.	Transaction of data in Augmented Reality is a type of (A) Unstructured Data (B) Structured Data (C) Semi Structured Data	1
2.	In how many forms Big Data could be found? (A) 3 (B) 5 (C) 4 (D) 2	1
3.	How many V's are there in BIG DATA (A) 5 (B) 4 (C) 3 (D) 6	1
4.	Data in _____ bytes size is called Big Data. (A)Tera (B)Giga (C)Peta (D) Meta	1
5.	The "Amazon EC2" can serve as a practically unlimited set of _____ machines. (A) Virtual (B) Real (C) Distributed	1

6.	In a Hadoop Cluster the <i>JobTracker</i> pushes work out to which of the available nodes striving to keep the work as close to the data as possible. (A) DataNodes (B) TaskTracker (C) ActionNodes	1
7.	Which of the following serves as the master and there is only one NameNode per cluster. (A)Data Node (B)NameNode (C)Data block (D)Replication	1
8.	The NameNode will interactively prompt you at the command line about possible courses of action you can take to recover your data in one of the following modes: (A) Full (B) Recovery (C) Partial (D) Commit	1
9.	The configuration file must be owned by the user running (A)DataManager (B)NodeManager (C)ValidationManager	1
10.	DataNode is the slave/worker node and holds the user data in the form of Data Blocks. Is the following statement TRUE?	1

	(A) TRUE (B) FALSE	
11.	In MapReduce Which of the following phases occur simultaneously? (A) Sort and Shuffle (B) Reduce and Sort (C) Shuffle and Map	1
12.	The Mapper implementation processes one line at time using the method: (A)Map (B)Mapper (C)Reduce (D)Reducer	1
13.	The primary interface for a user to describe a MapReduce job to the Hadoop framework for execution. (A) JobConf (B) MemoryConf (C) Map Parameters	1
14.	For which of the following does the MapReduce framework spawn one map task generated by the <i>InputFormat</i> for the job. (A) InputSplit (B) OutputSplit (C) InputSplitStream	1
15.	Which of the following is true for RecordReader <K, V> in MapReduce as it assumes the responsibility of processing record boundaries and presenting the tasks with keys and values. (A) It reads <key, value> pairs from an InputSplit for	1

	only mapper task to process (B) It reads <key, value> pairs from an InputSplit for only reducer task to process (C) It reads <key, value> pairs from an InputSplit and presents a record-oriented view for the Mapper and Reducer tasks for processing	
16.	HBase is _____ defines only column families. (A) Fixed Schema (B) Schema-less (C) Row Oriented	1
17.	The _____ Server assigns regions to the region servers and takes the help of Apache ZooKeeper for this task. (A) Region (B) Master (C) Zookeeper	1
18.	HBase Admin and _____ are the two important classes in this package that provide DDL functionalities. (A) HDescriptor (B) HTableDescriptor (C) HTable (D) HTabDescriptor	1
19.	You can delete a column family from a table using the method _____ of HBaseAdmin class. (A) delColumn() (B) deleteColumn() (C) removeColumn()	1

20.	The main configuration file of HBase is: (A) hbase.xml (B) hbase-site.xml (C) hbase-site-conf.xml	1
21.	The figure below is an Airflow DAG. Please select the right answers to identify the dependencies of the following Spark Job submitted using SparkSubmitOperator operator.  (A) The flight_search_waiting_time and flight_nb_search run in parallel if the first task flight_search_ingestion is done successfully. (B) The flight_nb_search runs only after flight_search_waiting_time runs successfully after first task flight_search_ingestion (C) The number of task instances across the DAG to define parallelism is 2	1
22.	Which of the following statement is true for the Apache Airflow (i) Creating and maintaining a relationship between tasks in cron is a challenge, whereas it is as simple as writing Python code in Airflow (ii) Executor is another internal component of the scheduler in Airflow (iii) Cron jobs are reproducible as they are configured externally. Airflow maintains an audit trail of all tasks completed	1

	(A) Both (i) and (iii) are TRUE (B) Both (ii) and (iii) are TRUE (C) Both (i) and (ii) are TRUE	
23.	Which will connect the 2 nodes in the figure below amongst these?  (A) Zookeeper (B) Broker (C) Server (D) Message	1
24.	A replication is necessary in Kafka because: (A) A published message will not be saved (B) A published message will not be lost (C) A published message will not be sent (D) A published message will not be deleted	1
25.	A Kafka Cluster has 5 brokers for which 2000 topics were created with 50 partitions each and 1 replica how many partitions will be allowed for the cluster: (A) 100,000 (B) 10,000,000 (C) 10,001	1

26.	The results of a HQL(Hive Query) can be stored on: (A) HDFS (B) Local (C) HDFS and LOCAL files	1
27.	The partitioning of tables in HIVE creates which of the following: (A) Files under the table name (B) Files under database name (C) Subdirectories under the table name (D) Subdirectories under the database name	1
28.	For logging HIVE uses : (A) logj4 (B) log4l (C) log4j (D) log4i	1
29.	Are the following statements TRUE? (i) Spark enables Apache HIVE users to run their unmodified queries much faster (ii) HIVE connects with SPARK SQL based on following steps : 1. Put <i>hive-site.xml</i> on your classpath, and specify <i>hive.metastore.uris</i> to where your hive metastore hosted. 2. Import <i>org.apache.spark.sql.hive.HiveContext</i>, as it can perform SQL query over Hive tables. 3. Define <code>val sqlContext = new org.apache.spark.sql.hive.HiveContext(sc)</code>. 4. Verify <i>sqlContext.sql("show tables")</i> to see if it works?. Choose from the following options: (A) Both (i) and (ii) are TRUE (B) Only (i) is TRUE (C) Only (ii) is TRUE	1

30.	Can the name of a view be similar to the HIVE Table name (A) YES (B) No	1
31.	If you set the <i>Inferschema</i> to be "false". The attributes of the dataframe will automatically casted to which of the following data types? (A) Integer (B) Double (C) Compilation Error (D) String	1
32.	Spark Structured Streaming can be defined as follows: (A)As Spark Structured Streaming has an underlying table, its functions/operations are similar to batch processing. (B)Spark Structured Streaming treats live data as a table that is continuously appended. (C)Spark Structured Streaming API can be used to process graph data .	1
33.	DStreams was built on Resilient Distributed Datasets (RDDs). Real streaming never existed in the case of DStreams. Spark structured streaming works on the concept of micro-batches. Which of the following is valid that Structured Spark Streaming eliminated (A)The lateness of data was not handled properly (B)Fault tolerance was not in place for driver programs. (c)Memory optimisations were also missing since RDDs are not a data structure.	1

34.	Given a dataframe df, select the code that returns its number of rows (A) df.take('all') (B) df.count() (C) df.collect() (D) df.numRows()	1
35.	The function that returns true for the transformation returning a new DStream by selecting only those records of the source DStream is: (A) transform (func) (B) reduceByKey() (C) map(func) (D) filter (func) (E) flatmap()	1
36.	Fault Tolerance in RDD is achieved using (A)Immutable nature of RDD (B)Lazy-evaluation (C)DAG (Directed Acyclic Graph) (D)Persistence	1
37.	The following statement is correct: Spark Streaming does not support the core Spark Concepts such as Lazy Evaluation, Transformation and Actions: (A) TRUE (B) FALSE	1
38.	The shuffling of data will not be generated from any executor across the cluster by which of the following DataFrame command? (A)df.orderBy() (B)df.repartition() (C)df.distinct()	1

	(D)df.map()	
39.	The following code : <pre> from pyspark.ml.feature import VectorIndexer from pyspark.mllib.util import MLUtils data = MLUtils.loadLibSVMFile(sc, "data/mllib/sample_libsvm_data.txt").toDF() indexer =VectorIndexer(inputCol="features", outputCol="indexed", maxCategories=10) indexerModel = indexer.fit(data) # Create new column "indexed" with categorical values transformed to indices indexedData = indexerModel.transform(data) </pre> specifies which of the features of Spark MLLib (A)Extraction: Extracting features from “raw” data (B)Transformation: Scaling, converting, or modifying features (C)Selection: Selecting a subset from a larger set of features	1
40.	Spark MLLIB supports two types of vectors: dense and sparse. Now for the vector: vector(1.0, 0.0, 3.0) The dense vector is :[1.0,0.0,3.0] for Sparse Vector which one is correct: (A) [4, (0,2),(1.0,3.0)] (B) [3, (0,2) , (1.0,3.0)] (C) [3,(1,2), (0.0, 3.0)]	1